# Final Project
# Group 35

BUSN 32120

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# Introduction:

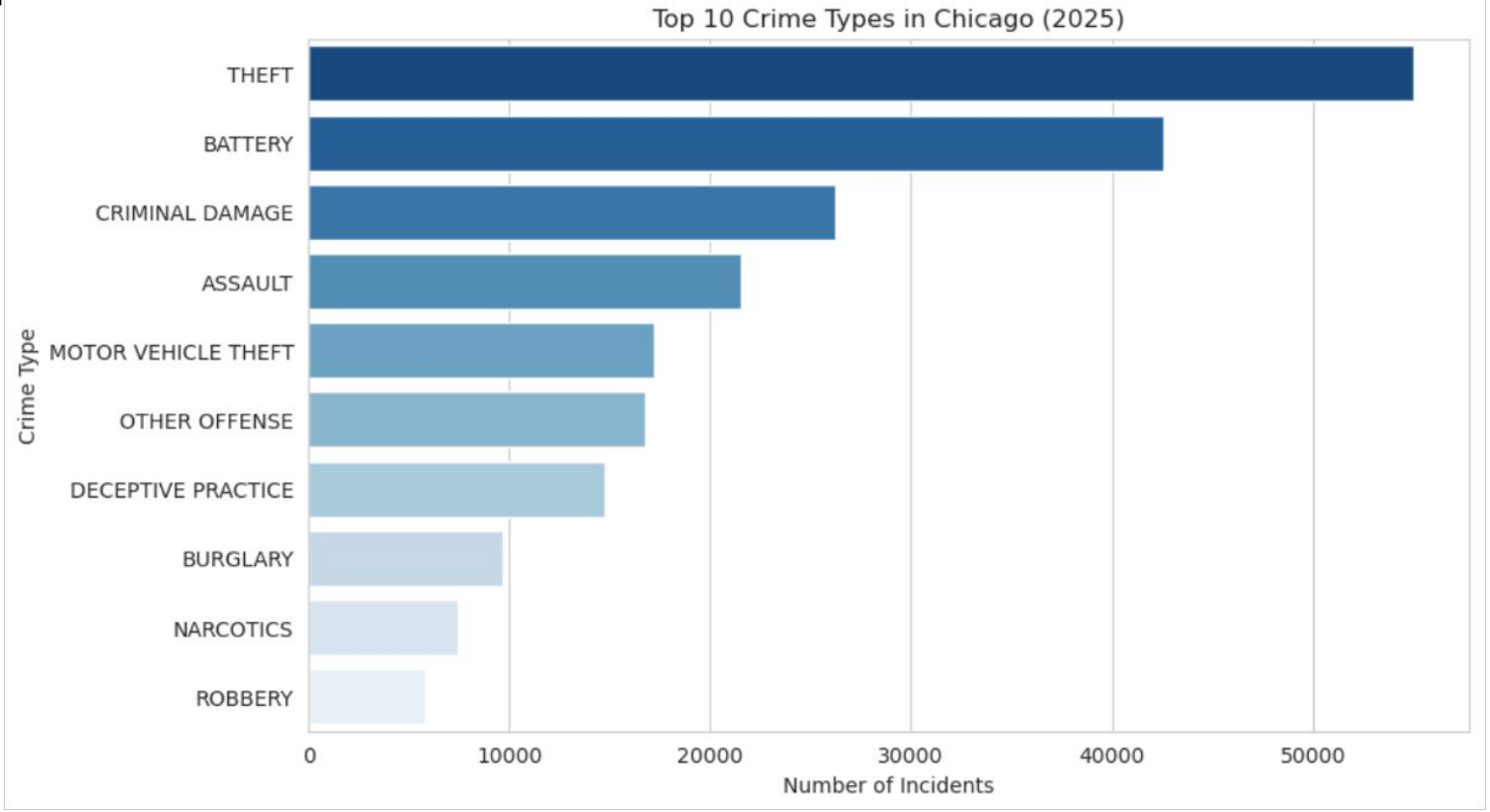
**Target Audience:**

- This analysis of 2025 Chicago crime data is intended for the public to make informed decisions regarding crime patterns across the community.

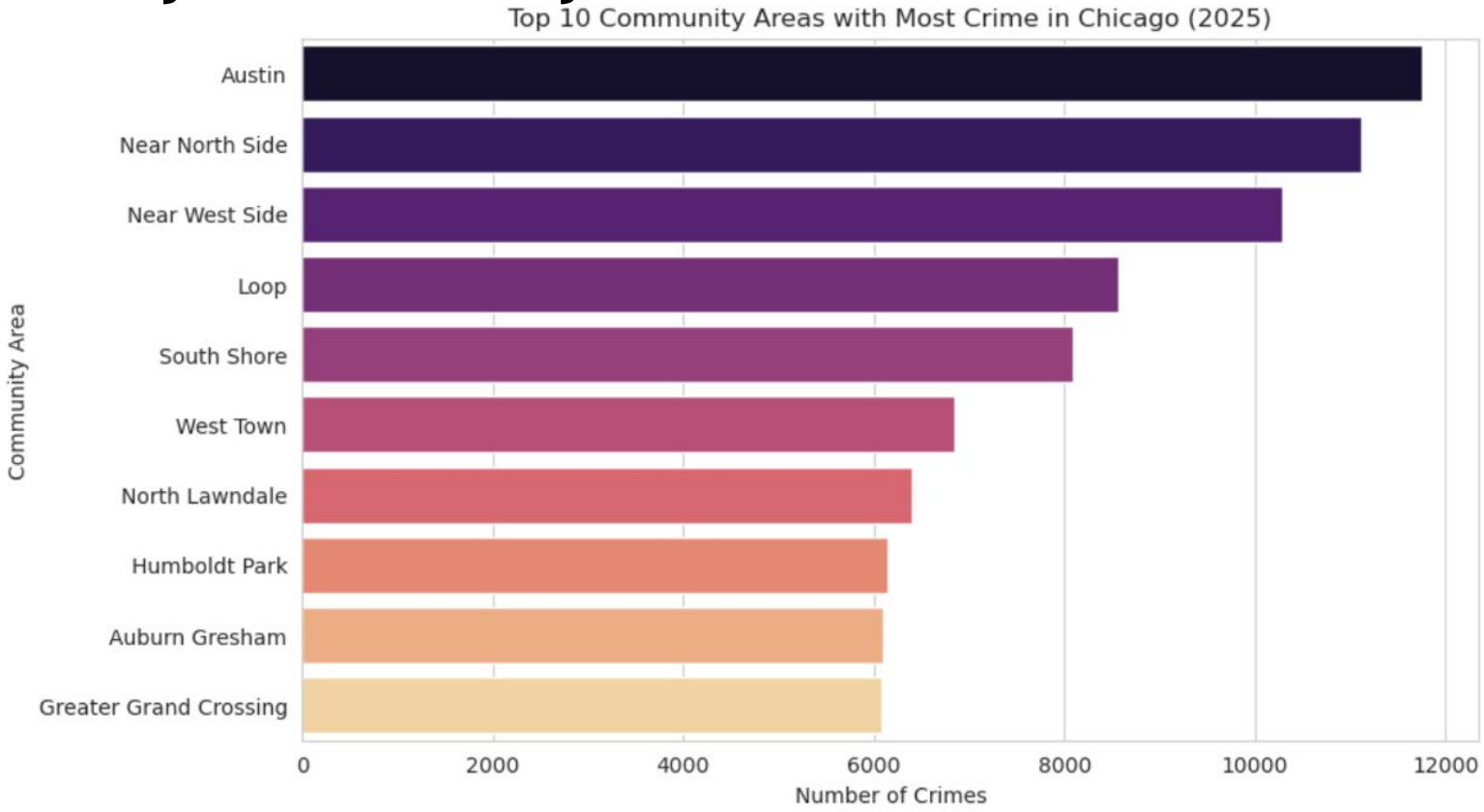
**Relevance and Motivation:**

- This analysis focuses on crime trends by type, location and time to help identify areas needing targeted interventions.
- The goals is to provide actionable insights to improve safety.

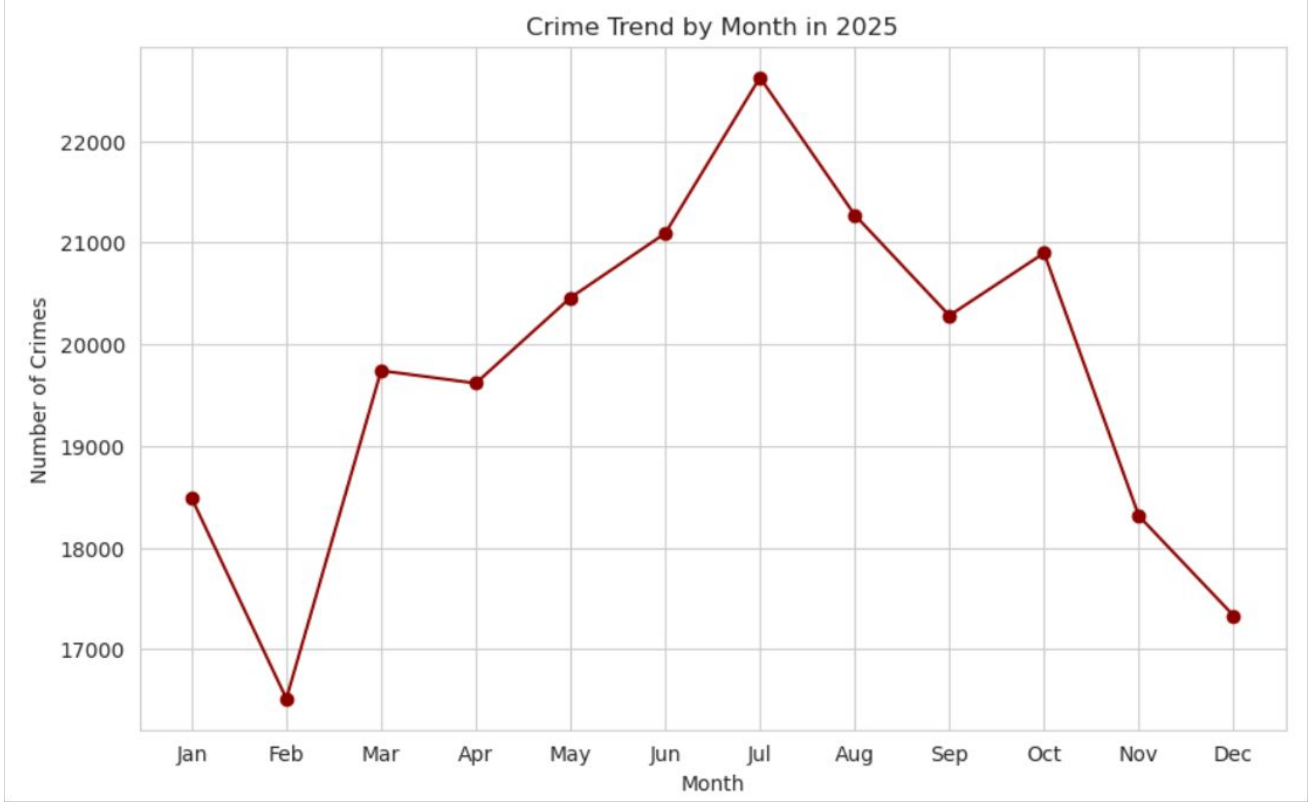
# Top Crime Types



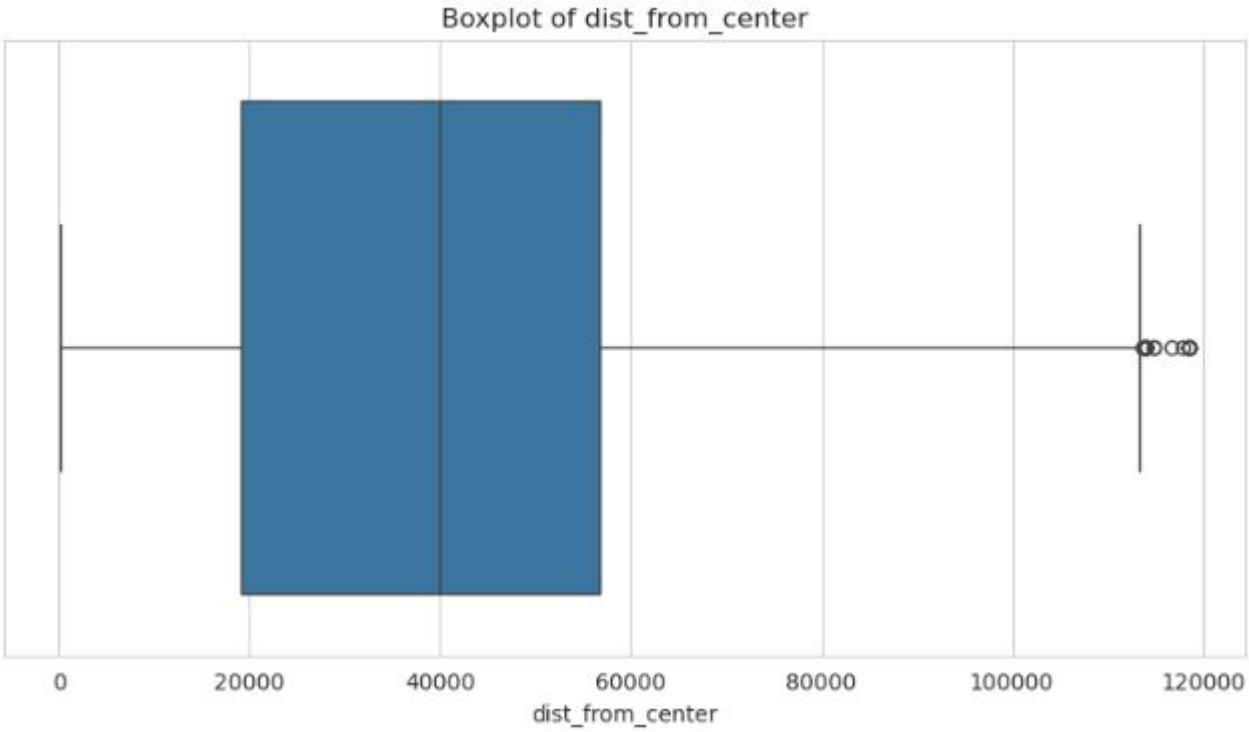
# Crimes by Community Area



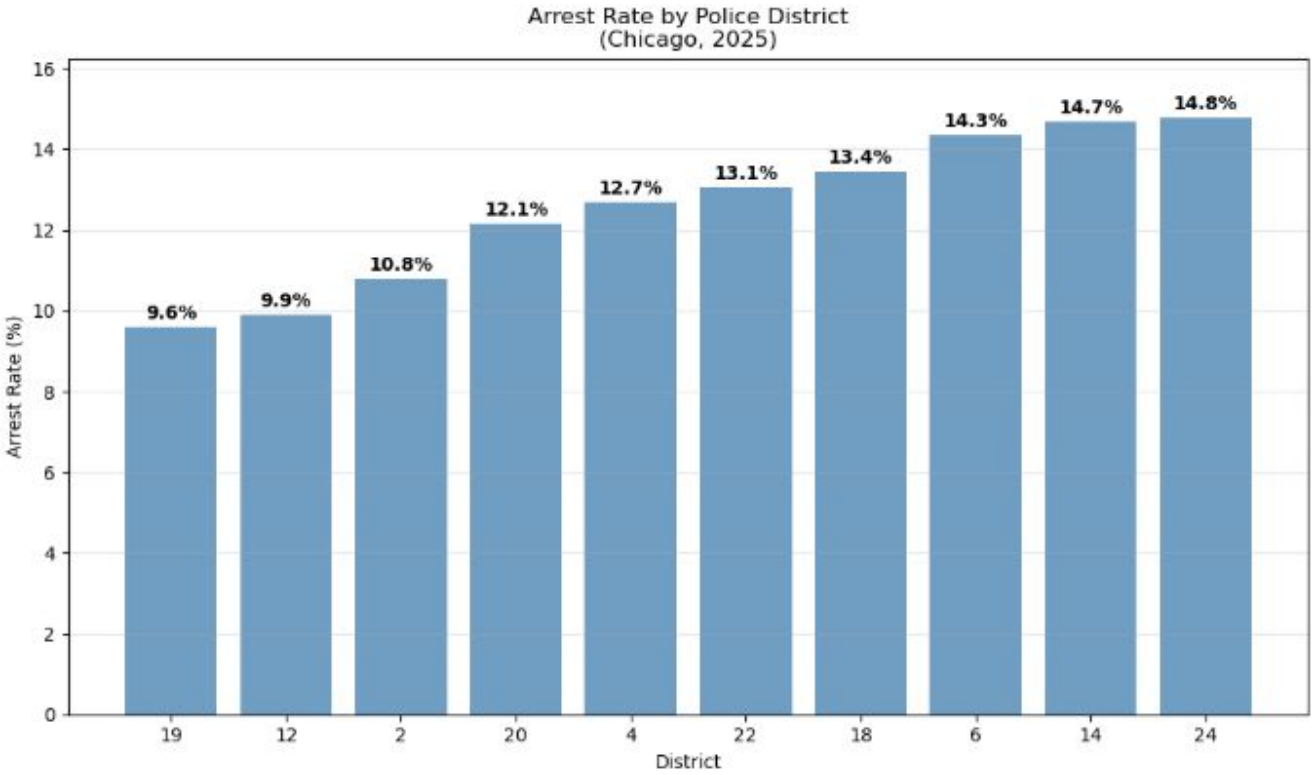
# Crime Trend by Month



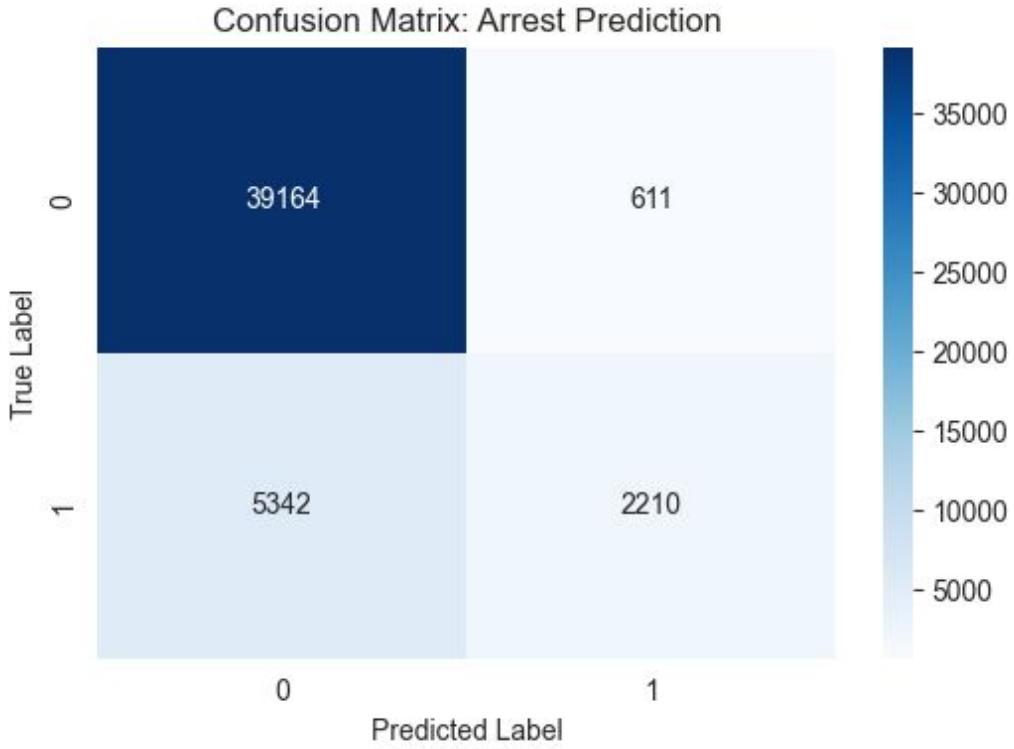
# Crimes Distance from the Center of Chicago



# District Arrest Rates



# Can a linear model predict whether a crime resulted in an arrest?

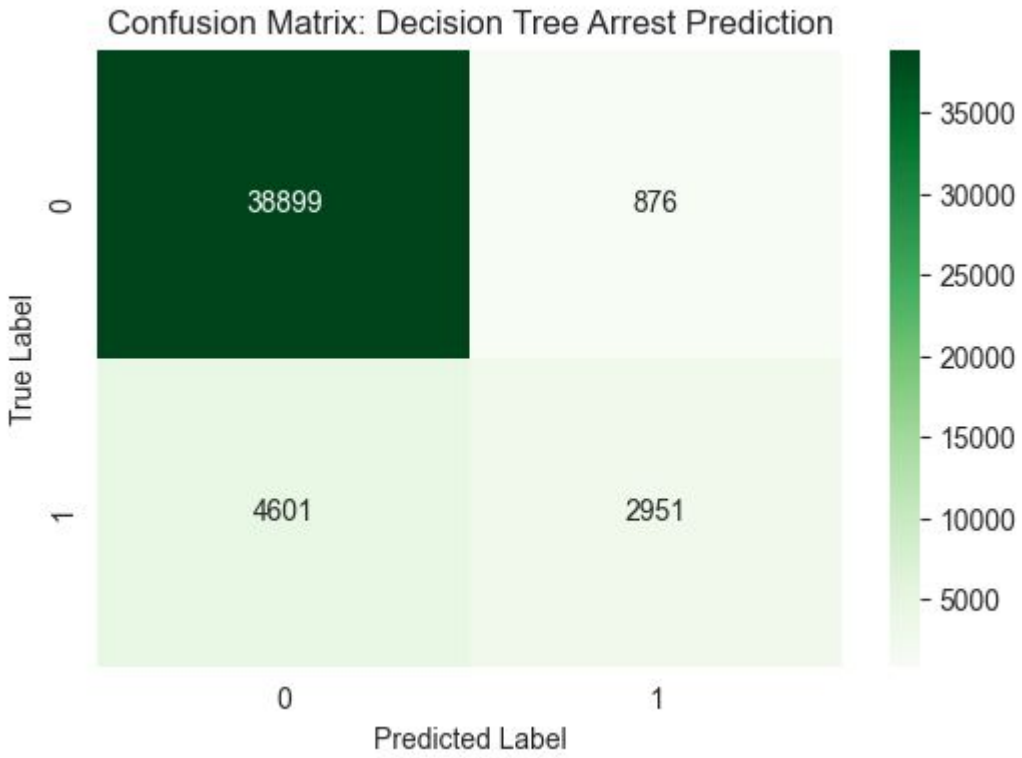



**Logistic Regression Performance:**

- Accuracy: 0.8742
- Precision: 0.7834
- Recall: 0.2926
- F1 Score: 0.4261
- ROC-AUC: 0.8184

- The model was found to be relatively conservative in predicting arrests. It avoids many false positives, but at the cost of missing many true arrest outcomes

# Can a nonlinear model predict whether a crime resulted in an arrest?



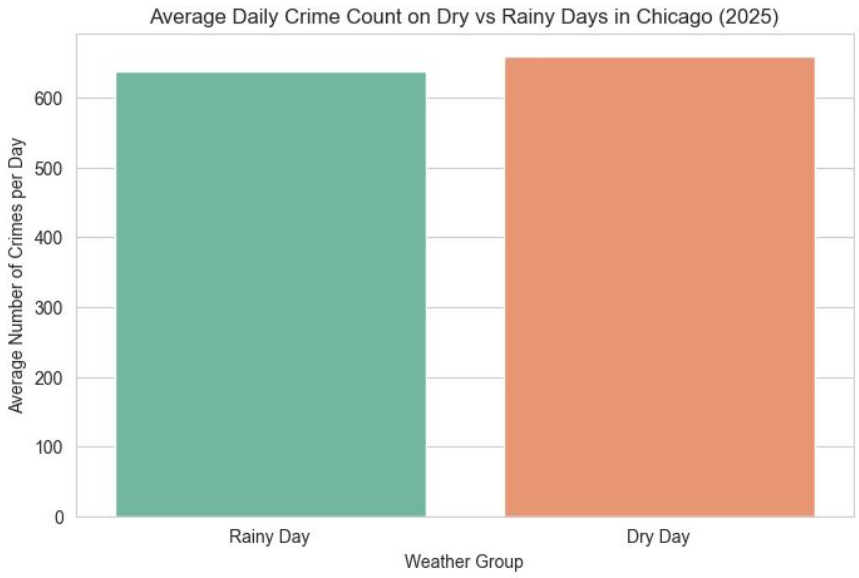

**Decision Tree Performance:**

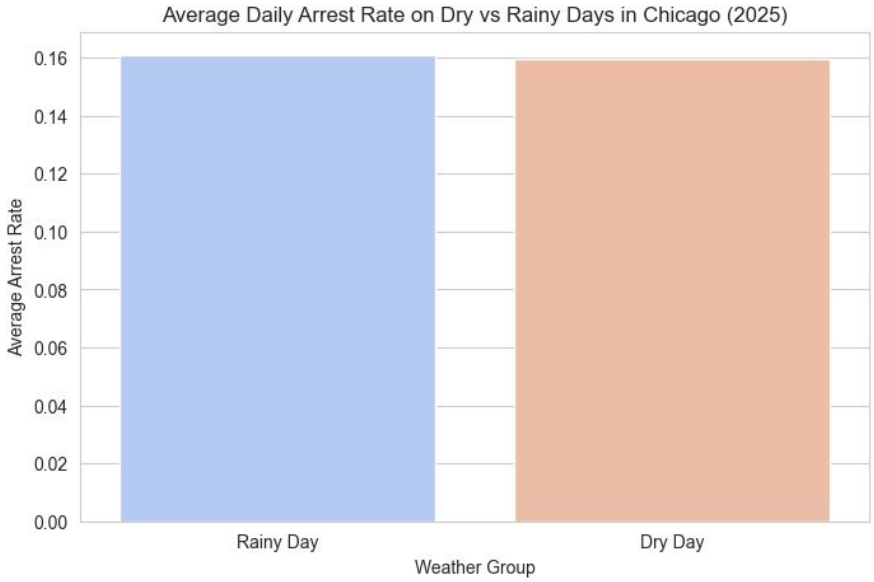
- Accuracy: 0.8843
- Precision: 0.7711
- Recall: 0.3908
- F1 Score: 0.5187
- ROC-AUC: 0.8361

- The model correctly identified 38,899 non-arrest cases and 2,951 arrest cases, while missing 4,601 arrest cases.

- Recall was 0.3908, which was higher than in the logistic regression model.

# Does daily weather relate to crime volume and arrest outcomes?



# Conclusion: What did we learn about Chicago crime in 2025?

**EDA and Features on Crime**

- Crime and arrest outcomes varied across offense type, time, and geography.
- Arrests were more predictable when using crime, district, and time-based features together.

**Model on Crime, Crime + Weather**

- The nonlinear decision tree captured these patterns slightly better than logistic regression.
- Weather added context, but had only a modest relationship with daily crime volume and little effect on arrest rate.

- These findings can help support more informed discussions about public safety and resource allocation.